

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
1.	(a)	(i)		Ovulation	1			1		
		(ii)		Any four (×1) from A. (the acrosome reaction) releases {hydrolases/(named) enzymes} which {digest/ break down} the zona pellucida (1) Ignore refs to membranes B. the {genetic material/ DNA/ chromosomes} of the sperm cell enters (the secondary oocyte) (1) C. this triggers {the cortical reaction/ formation of the fertilisation membrane/ or description of} and prevents {polyspermy/ or description of} (1) D. (entry of the genetic material also) triggers meiosis II (1) E. (nuclei of the sperm and ovum/ gametes fuse to) form a <u>diploid</u> {zygote/nucleus} (1)	4			4		
		(iii)		<u>Mitosis</u> + {cleavage/ cytokinesis}	1			1		
	(b)			{Cilia/ they} move / owtte (1) the {secondary oocyte/ zygote/blastocyst} (1) (from the ovary) along the {fallopian tube/ oviduct} towards the {uterus/ endometrium} (1)		3		3		1
	(c)	(i)		Any two (×1) from: (The length of gestation/ 5 months for ewe/ almost a year for mare/ it) means offspring born in spring (1) means that {fresh grass/ more food} will be available for the mother (better nutrition for milk production) / {lamb/foal} will have {grass/ food} in spring (1) Warmer temperatures/not winter/cold / ORA(1)			2	2		

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		(ii)		Any two (×1) from: More likely to result in fertilisation (1) Sperm need time to swim (to top of/ along) fallopian tube(1) May not mate until later in oestrus (1)			2	2		
				Question 1 total	6	3	4	13	0	1